

Surakiat Kansa-ard

Junior AI Engineer

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 [surakiat-kansa-ard](#)  <https://github.com/SurakiatP>

SKILLS

Programming: Python, SQL, C/C++

ML/DL Frameworks: PyTorch, TensorFlow, scikit-learn, pandas, numpy

Generative AI & LLMs: LangChain, Hugging Face, OpenAI API, Ollama, RAG Systems, Prompt Engineering, NLP

Computer Vision: OpenCV, CLIP, OCR (EasyOCR, Tesseract), Image Classification

Vector Databases: FAISS, Qdrant, Chroma | **Embeddings:** BGE-m3, MPNet, multilingual-e5

Data Engineering: ETL Pipelines, Airflow, Kafka, Data Cleaning, PyThaiNLP

Backend & Deployment: FastAPI, Docker, REST API, CI/CD (GitHub Actions)

MLOps: MLflow, DVC, Model Versioning, Experiment Tracking

Visualization: Streamlit, Gradio, Matplotlib, Plotly, Grafana

Cloud: AWS (exposure), GCP (exposure) | **Version Control & Containerization:** Git, GitHub, Docker

Personal Traits : Curious, Collaborative, Eager to learn, Strong problem-solving, Willingness to explore new technologies

EDUCATION

King Mongkut's Institute of Technology Ladkrabang

Bachelor of Engineering | Major: Automation Engineer | Cumulative GPA: 3.30/4.00

PROFESSIONAL EXPERIENCE

IoT Solution Engineer Intern

Swift Dynamics (Bangkok, TH)

May 2023 - July 2023

Contributed to R&D team developing Smart Camera Security solutions for manufacturing and industrial applications, implementing computer vision prototypes for real-time surveillance systems.

Accomplishments:

- Developed real-time anomaly detection system using OpenCV and computer vision algorithms on Linux platforms (Ubuntu, Debian), reducing human monitoring errors by 75% through automated surveillance.
- Implemented 24/7 automated security monitoring pipeline processing 50+ camera feeds simultaneously, detecting suspicious activities with 88% accuracy and alerting security personnel in real-time.

PROJECTS

Multi-modal RAG Shop Assistant (LLMs + Multi-RAG) | <https://github.com/SurakiatPI/Multi-modal-RAG>

End-to-end Multi-modal RAG system enabling product search through text, voice, and image queries with CLIP embeddings and optimized retrieval pipelines.

- Architected multi-modal RAG pipeline achieving 86.7% precision, 93.3% recall, and <1s latency using CLIP (ViT-B/32) for image-text embeddings, Sentence-Transformers (all-MiniLM-L6-v2) for semantic search, and Qdrant vector database processing 500+ products.
- Engineered hybrid retrieval combining dense vector search with BM25 keyword matching, implementing intelligent chunking (chunk_size=500, overlap=100) and metadata filtering for optimal context retrieval with automatic source citation.
- Integrated OpenAI GPT-4 API with advanced prompt engineering (few-shot learning, chain-of-thought reasoning), Whisper API for voice input, and PyThaiNLP for Thai language processing with edge-TTS multilingual synthesis.
- Deployed Streamlit web application with async FastAPI backend, custom evaluation framework (MRR, NDCG metrics), comprehensive error handling, and production monitoring with response time tracking.

Technologies: Python, OpenAI API, CLIP, Qdrant, Sentence-Transformers, LangChain, Streamlit, FastAPI, RAG, PyThaiNLP, Whisper, Computer Vision, NLP

Multi-Stage Retrieval RAG System (Hybrid Retrieval + Bilingual) | <https://github.com/SurakiatPI/Multi-Stage-Retrieval-RAG>

End-to-end bilingual (Thai/English) Retrieval-Augmented Generation system for corporate policy documents featuring hybrid search, cross-encoder reranking, and temporal conflict resolution.

- Developed end-to-end RAG pipeline achieving 96.7% evaluation accuracy (97.5% Faithfulness, 98.5% Answer Relevancy, 98.0% Policy Compliance) using DeepEval framework with LLM-as-a-Judge (Groq llama-3.3-70b) on 10 test cases covering policy updates and conflict resolution.
- Architected hybrid retrieval system combining BM25 sparse search (40%) with FAISS dense vector search (60%) using HuggingFace multilingual-e5-small embeddings, followed by BAAI/bge-reranker-base cross-encoder reranking (Top-5), processing 1,200+ document chunks with sub-2s query latency.
- Engineered bilingual document processing pipeline: Thai documents via EasyOCR rasterization + PyThaiNLP normalization/tokenization, English documents via Docling extraction, with RecursiveCharacterTextSplitter (chunk_size=1200, overlap=300) for semantic preservation.
- Implemented LLM-powered query expansion using scb10x/typhoon2.1-gemma3-4b generating Thai-English keyword pairs, and context-aware answer generation using qwen2.5:7b-instruct with TOON tabular format for structured context and automatic source citation.
- Deployed production FastAPI REST API with async endpoints, comprehensive logging, and Ollama integration for local LLM inference, reducing operational costs while maintaining <2s end-to-end response time.

Technologies: Python, LangChain, FAISS, Hugging Face Transformers, Ollama, PyTorch, FastAPI, PyThaiNLP, EasyOCR, DeepEval, Docker, RAG, NLP, Semantic Search, TOON Format

Uber Delivery Time Prediction MLOps (PoC to Demo MLOps Pipeline) | <https://github.com/SurakiatPI/uber-delivery-time-prediction-mlops>

End-to-end MLOps pipeline for real-time delivery time prediction with automated retraining, streaming inference, and production monitoring using Kafka, MLflow, Airflow, and Prometheus/Grafana.

- Architected production MLOps pipeline achieving 60% faster training time using XGBoost (MAE: 3.2min, RMSE: 4.8min) with automated hyperparameter tuning (RandomizedSearchCV, n_iter=10, cv=3), experiment tracking via MLflow, and model versioning with DVC for reproducibility across 1,155 delivery records.
- Engineered feature pipeline processing weather (OpenWeatherMap API), traffic, and temporal features with 25+ engineered variables including distance binning (4 categories), hour buckets (4 periods), severity mapping (weather: 0-3, traffic: 0-2), and one-hot encoding for categorical variables.
- Implemented real-time streaming inference using Kafka (2s intervals), consuming order events with FastAPI serving predictions (<100ms latency) and maintaining 95%+ prediction success rate with automatic model loading from MLflow registry (Production stage).
- Deployed automated daily retraining pipeline using Apache Airflow with DVC integration for data versioning, PostgreSQL backend for metadata management, and MLflow model registration ensuring continuous model improvement with zero downtime deployment.
- Established production monitoring stack with Prometheus metrics (prediction latency, success/failure rates, input distributions) and Grafana dashboards tracking system health, model performance, and resource utilization across containerized microservices (Docker Compose: 9 services).

Technologies: Python, Machine Learning, XGBoost, LightGBM, MLflow, Apache Airflow, DVC, Kafka, FastAPI, Docker, PostgreSQL, Prometheus, Grafana, Scikit-learn, Pandas, NumPy, MLOps, CI/CD

CERTIFICATIONS

DataCamp Machine Learning Engineer : [DataCamp ML Engineer Certificate](#)

Skills: MLOps, End-to-End ML Pipeline, MLflow, Data Version Control (DVC), CI/CD, Shell

DataCamp Associate Data Scientist in Python : [DataCamp Associate Data Scientist in Python Certificate](#)

Skills: Python, Pandas, NumPy, Scikit-learn, Statistics, Matplotlib, Seaborn, Exploratory Data Analysis, Data Cleaning, Supervised/Unsupervised Learning

Codecademy ML/AI Engineer Career Path : [Codecademy ML /AI Engineer Career Path Certificate](#)

Skills: Reinforcement Learning, Machine Learning Algorithms, Scikit-learn